



OBJECT RECOGNITION And DISTANCE MEASUREMENT

Using Raspberry Pi 5 And Hailo-8L AI

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Imagine a vehicle cruising at 54 kilometres per hour, roughly the speed of an object moving 0.5 metres per frame at 30 frames per second ($0.5 \times 30 \times 3.6 = 54 \text{ km/hr}$). Mounted on the vehicle's roof is a compact, high-performance vision system built around a Raspberry Pi 5 board and a Hailo-8L AI accelerator. This real-time setup detects objects ahead and estimates their distances from the vehicle. When an object enters a predefined safety zone, the system can instantly issue warnings or activate emergency braking, enhancing both situational awareness and reaction time.

Built using an AI Accelerator HAT on a Raspberry Pi 5, the system operates as a production-grade AI platform capable of running at a minimum of 30 frames per second. Designed for vehicle mounting, it continuously scans the road, identifies objects in real time, and calculates their distances. These live readings are displayed on the terminal. If any object crosses a predefined proximity threshold, the system may trigger alerts via GPIO pins. Although the alert mechanism is straightforward to implement, it remains unbuilt in the current version. The components required for assembling the system are listed in the Bill of Materials table. Fig. 1 shows the authors' working prototype.

Connection and mounting

The AI accelerator can be mounted

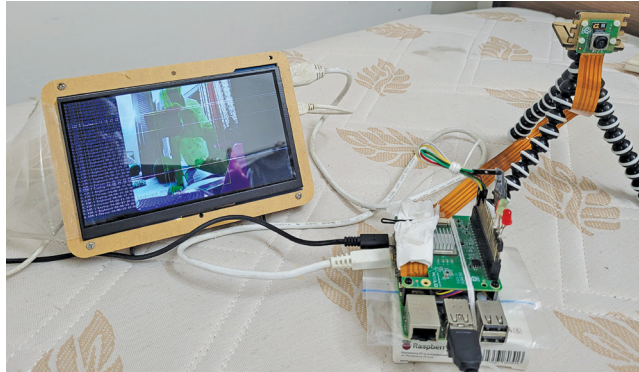


Fig. 1: Authors' prototype

BILL OF MATERIALS

Components	Quantity
Raspberry Pi 5, 8GB or 16GB	1
Hailo-8L AI accelerator for Pi 5, 13 TOPS to 26 TOPS	1
Pi 5 camera	1
Pi 5 power adaptor	1
High-speed SD card, 32GB or more	
HDMI adaptor/display	1

on top of the Raspberry Pi, as shown in Fig. 2. Note that since the AI HAT fully covers the main processor, the Raspberry Pi 5 tends to heat up significantly. To prevent overheating, it is strongly recommended to use an additional cooling method. In this system, only a specific fixed-type cooler fits within the narrow space available once the AI HAT is mounted. The complete setup and its individual components are shown in Fig. 2. The Raspberry Pi 5 must also be powered by a high-quality 5V, 4A power adaptor to maintain stable performance.

With active cooling, the Raspberry Pi 5 can be safely over-clocked to extract additional performance, at the cost of slightly increased power consumption and heat generation. The cooler's fan connector should be plugged into

the designated fan header on the Pi 5 board. When paired with the AI HAT, the heavy lifting of YOLO-based AI processing is handled by the accelerator, allowing the Pi 5 to focus on computations and control logic. This division of labour ensures smooth and efficient operation, even at 30 frames per second.

Software

First, ensure that 64-bit Raspberry Pi OS Bookworm is installed. This system does not function on a 32-bit Pi OS. A high-quality Camera Module 3 or better is recommended for best results.

Installation guide. <https://www.raspberrypi.com/documentation/computers/ai.html#getting-started>
Update the Systembash

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

Accessing Raspberry Pi. The most convenient method for operating the Raspberry Pi is by accessing it from a desktop or laptop via SSH: